My second explanation is tsunamis. As Miller's planet rocks, Gargantua's tidal forces may not pulverize its crust, but they do deform the crust first this way and then that, once an hour, and those deformations could easily produce gigantic earthquakes (or "millerquakes," I suppose we should call them). And those millerquakes could generate tsunamis on the planet's oceans, far larger than any tsunami ever seen on Earth, such as the one that hit Miyako City, Japan, on March 11, 2011 (bottom half of Figure 17.7).

Past History of Miller's Planet

It is interesting to speculate about the past and future history of Miller's planet. Try it using as much physics as you know or can scrounge up from the web or elsewhere. (This is not easy!) Here are some things you might think about.

How old is Miller's planet? If, as an extreme hypothesis, it was born in its present orbit when its galaxy was very young (about 12 billion years ago), and Gargantua has had its same ultrafast spin ever since, then the planet's age is about 12 billion years divided by 60,000 (the slowing of time on the planet): 200,000 years. This is awfully young compared to most geological processes on Earth. Could Miller's planet be that young and look like it looks? Could the planet develop its oceans and oxygen-rich atmosphere that quickly? If not, how could the planet have formed elsewhere and gotten moved to this orbit, so close to Gargantua?

How long can the planet's rocking continue until friction inside the planet converts all the rocking energy to heat? And how long could it have rocked in the past? If a lot shorter than 200,000 years, then perhaps something set it rocking. What could have done so?